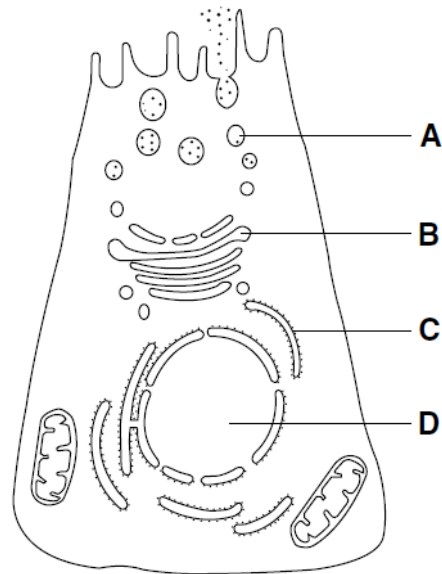


- 1 The diagram is taken from an electron micrograph of a cell which secretes digestive enzymes.

Where are these enzymes made?

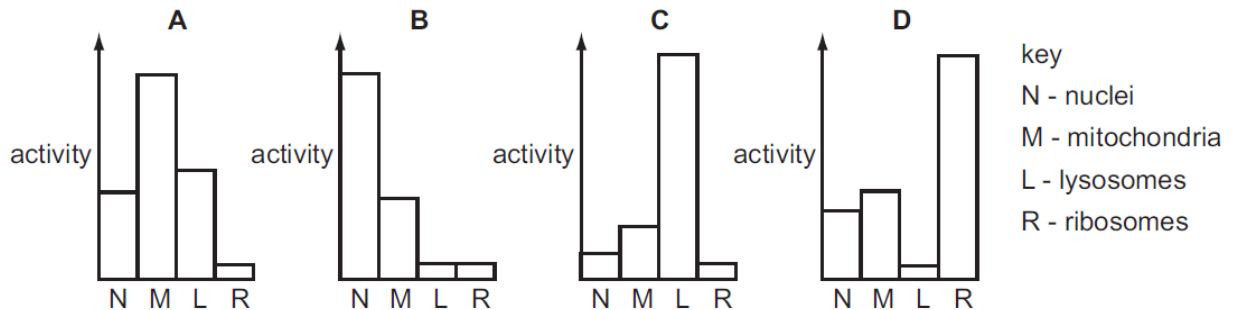


- 2 What is the order of size of cell components?

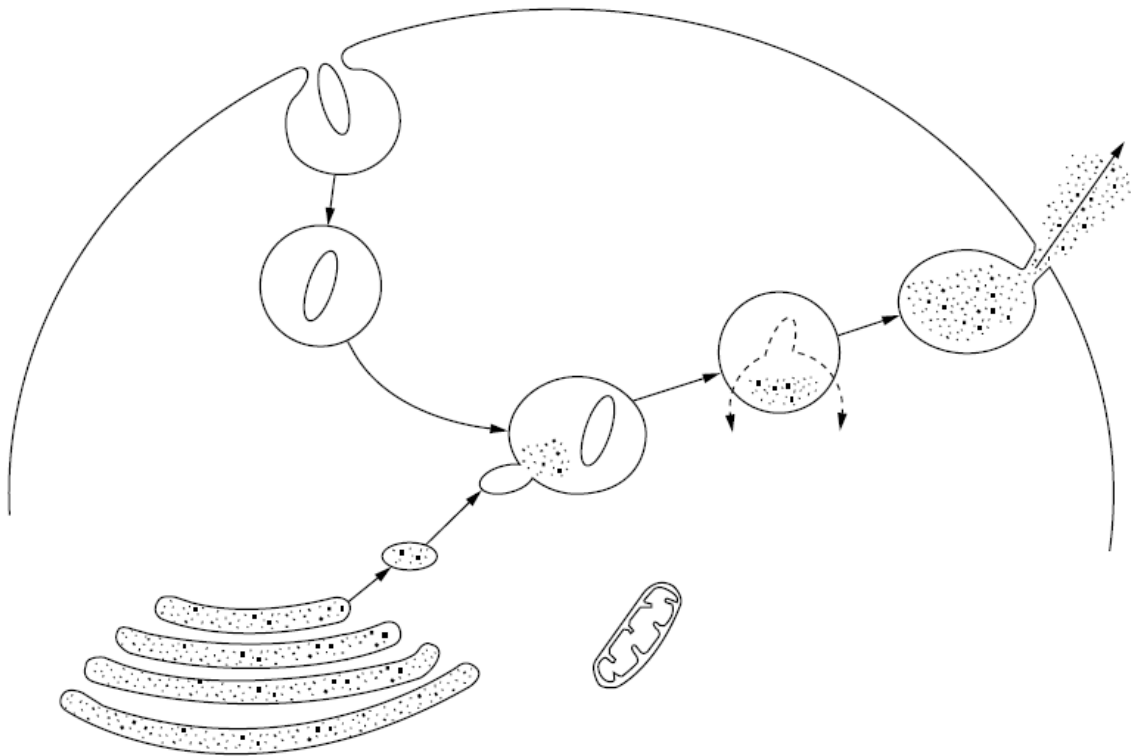
	Largest → Smallest			
A	Mitochondria	Ribosomes	Starch grains	Nuclei
B	Nuclei	Chloroplasts	Mitochondria	Ribosomes
C	Ribosomes	Mitochondria	Chloroplasts	Starch grains
D	Starch grains	Mitochondria	Chloroplasts	Ribosomes

- 3 A piece of mammalian tissue was homogenised and subjected to differential centrifugation to yield four subcellular fractions. The activity within each fraction, of four different types of enzyme, **A**, **B**, **C** and **D**, was investigated.

Which bar chart shows the results of investigating hydrolytic enzyme activity?



- 4 The diagram shows several processes taking place in a cell.



Which processes are shown in the diagram and involve the cell surface membrane of the cell?

- A active transport and diffusion
- B diffusion and osmosis
- C endocytosis and exocytosis
- D endocytosis and osmosis

- 5 A plant cell is placed in a solution with a less negative (higher) water potential than the cell contents.

Which change occurs in the cell and what causes the change?

	change	cause
A	cell becomes more flaccid	solution diffuses out of the cell
B	cell becomes more flaccid	water diffuses out of the cell
C	cell becomes more turgid	solution diffuses into cell
D	cell becomes more turgid	water diffuses into cell

- 6 Beetroot cells contain a water-soluble red pigment. Two test tubes were set up as described in the table.

tube 1	Pieces of washed raw beetroot in water
tube 2	Pieces of washed raw beetroot in water containing 3 drops of cyanide, a respiratory inhibitor.

After 30 minutes, the water in tube 2 contained a red pigment but the water in tube 1 did not.

Which of the following statements are false for tube 2?

- 1 Pigment molecules passed out and were replaced by cyanide.
 - 2 The cell membrane was unable to retain the red pigment.
 - 3 Water entered the tissue by osmosis and caused the cells to burst.
 - 4 Water passed out of the cells by osmosis and carried the soluble pigment with it.
 - 5 The same result will occur if ethanol was used instead of cyanide.
- A** 1 and 3 only
- B** 3 and 4 only
- C** 2 and 5 only
- D** 1, 3 and 4 only

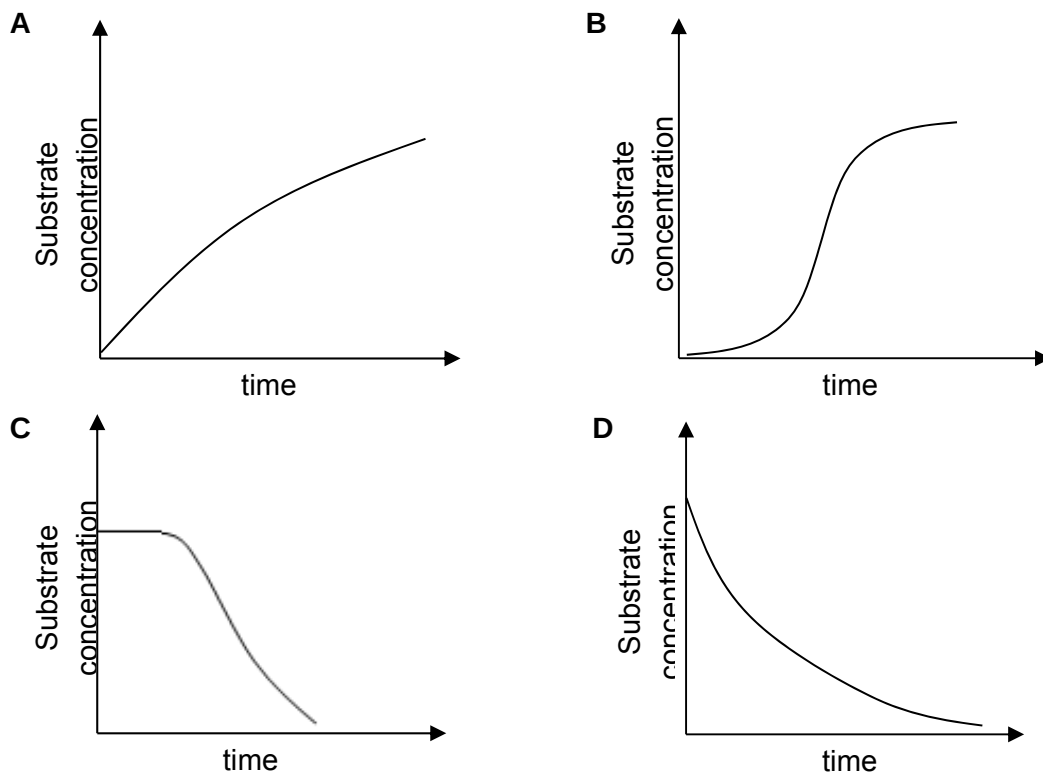
7 Some features of haemoglobin are listed.

- 1 They are made up of a protein component called globin and a non-protein component called haem group.
- 2 There are two types of polypeptide chains: the α -helix globin chain and the β -globin chain.
- 3 Each polypeptide chain is held by hydrophobic interactions, hydrogen and ionic bonds and disulfide bonds.
- 4 Most of its hydrophilic polar amino acid residues are on the external surface of the globular structure.

Which of these statements are true ?

- A** 1 and 3 **B** 1 and 2 **C** 1, 2 and 4 **D** 1 and 4

8 Which graph represents the changes in substrate concentration during the course of an enzyme catalysed reaction?



- 9** Which is a correct statement about obtaining human embryonic stem cells for research?
- A** Removal of these cells is considered to be ethically acceptable as normal development of the embryo is not inhibited.
 - B** The cells must be removed at an early stage of development from a region of the blastocyst known as the inner cell mass.
 - C** The cells must be removed within a day following the successful fertilisation of the ovum by the sperm, and after checking for normal mitotic division.
 - D** The region of the blastocyst from where the cells are removed is an area that develops at a later stage into the placenta.

- 10** Mesenchymal stem cells can differentiate into several types of cells belonging to our skeletal tissue, such as cartilage, bone and fat.

Which statement correctly describes mesenchymal stem cells?

- A** They are specialised cells that can give rise to a variety of cell types.
 - B** They can be stimulated by chemical signals to express certain genes.
 - C** They lose genetic information as they differentiate.
 - D** They occur in large numbers in the bone marrow.
- 11** Which statements about ribosomal RNA(rRNA) is true?
- 1 rRNA is involved in the binding of mRNA and not tRNA.
 - 2 rRNA with associated proteins, fold into complex tertiary structures within the ribosome.
 - 3 The 3' terminus of rRNA in prokaryotes is known to interact with the initiation region of mRNA via the Shine–Dalgarno sequence.
 - 4 Nucleolus is the site of rRNA transcription and processing, and of ribosome assembly.

- A** 1 and 3 **B** 1, 2 and 3 **C** 2 and 4 **D** 2, 3 and 4

12 What is the basis for the difference in the synthesis of the leading and lagging strands of DNA molecules?

- 1 The anti-parallel arrangement of the DNA strands.
- 2 The RNA primers are required to initiate DNA elongation.
- 3 DNA polymerase can join new nucleotides to the 3' end of the growing strand.
- 4 Helicase and single-stranded binding proteins work at the 5' end of the DNA strand.

A 2 and 4

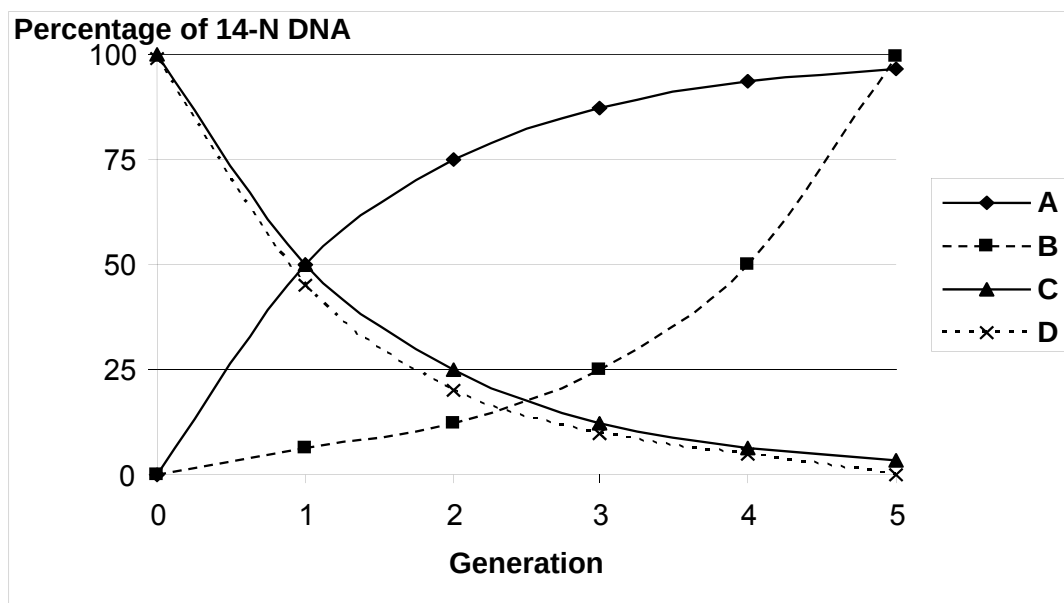
B 1 and 3

C 1 and 4

D 2 and 3

13 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for 5 generations. The percentage of ^{14}N DNA strands in each generation was estimated.

Which curve provides evidence that DNA replication is semi-conservative?



- 14** A new form of life is discovered. It has a genetic code much like that of organisms on Earth except that there are five different DNA bases instead of four and the base sequences are translated as doublets instead of triplets. How many different amino acids could be accommodated by this genetic code?
- A** 25
- B** 32
- C** 10
- D** 64
- 15** What do chromosomal aberrations and gene mutations have in common and how are they different?

	Similarity	Difference
A	Both may involve addition of nucleotides.	Gene mutations always produce dominant alleles but not chromosomal aberrations.
B	Both cause changes in number of chromosomes.	Gene mutations do not involve inversions but inversion of segments of chromosomes do occur.
C	Both affect DNA sequence.	Gene mutations occur within a chromosome but chromosomal aberrations may occur across chromosomes.
D	Both may not result in a difference in protein expression.	Chromosomal aberrations are more harmful than gene mutations.

- 16** The table shows the DNA triplet codes for some amino acids from the strand complementary to mRNA.

Amino acid	DNA triplet code
Glycine	CCA, CCG, CCT, CCC
Leucine	AAT, AAC, GAA, GAG, GAT, GAC
Lysine	TTT, TTC
Methionine	TAC
Proline	GGA, GGG, GGT, GGC
Threonine	TGA, TGG, TGT, TGC

The sequence of DNA triplets from the strand complementary to mRNA form part of a gene is shown.

...TAC TTT AAT GGC CCT GAG GGC TAC TGT ...

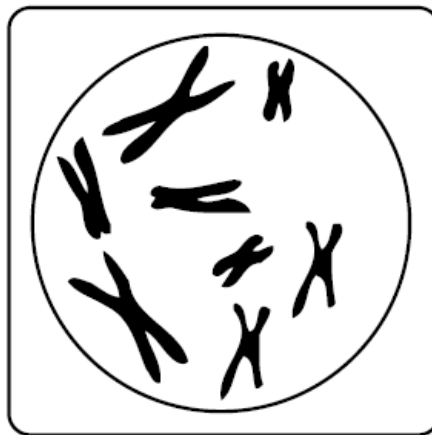
Which mutated DNA sequences of this part of a gene would result in the same amino acid sequences as the original gene sequences ?

- A** ...TAC TTT AAT GGC CCT GAG GGT CCA TGT
- B** ...TAC TTC GAT GGC CCT GAG GGC TAC TGT
- C** ...TAC TTT AAT GGC CCG GAG TGA TAC TGT
- D** TAC TTT AAT GGC CCT GAG GGC TTC TGT
- 17** Which environmental factor is associated with the development of melanomas (a type of skin cancer)?
- A** alcohol
- B** tar
- C** ultraviolet light
- D** X-radiation

- 18** Cancer research has found that gene amplification is involved in the development of cancer. This research has identified one type of sequence that causes a break in DNA before gene amplification.

How might the findings of this research be used?

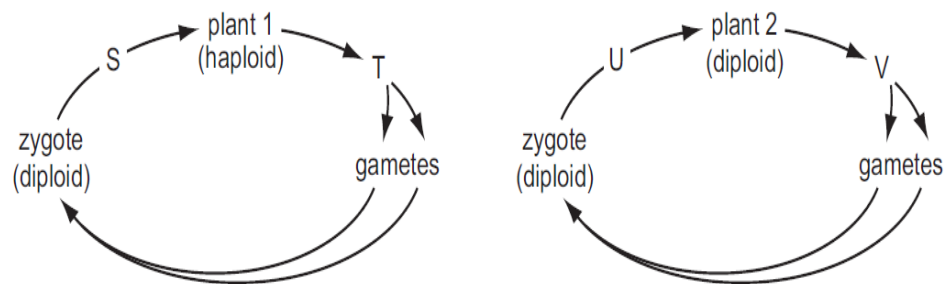
- A** to identify people at risk of developing cancer
 - B** to prevent cancer from developing in 'at-risk' people
 - C** to repair DNA breaks from amplification occurs
 - D** to use restriction endonuclease to remove DNA
- 19** The diagram shows a diploid cell during mitosis.



Which stage of mitosis is shown?

- A** anaphase
- B** metaphase
- C** prophase
- D** telophase

- 20 The diagram shows the life-cycles of two types of simple plant.



Where will reduction divisions occur in the life cycles?

- A at S and U
 B at S and V
 C at T and U
 D at T and V
- 21 In corn, the colour of the stem can be either wild type, which is dark green, or virescent, which is pale green.

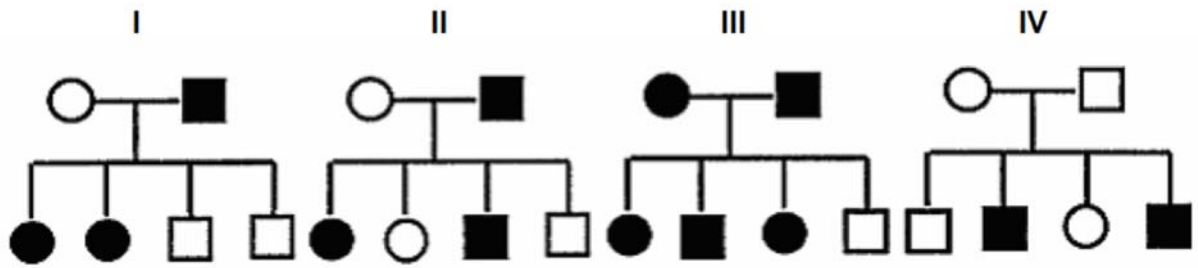
Cross	Phenotype of parents	Temperature at which raised	Offspring
1	virescent X virescent	20°C	virescent
2	virescent X virescent	37°C	wild type
3	wild type X wild type	20°C	3 wild type: 1 virescent
4	wild type X wild type	37°C	wild type

The parents of crosses 1 and 2 had the same genotypes. The parents of crosses 3 and 4 had the same genotypes.

Which of the following can be deduced from the data?

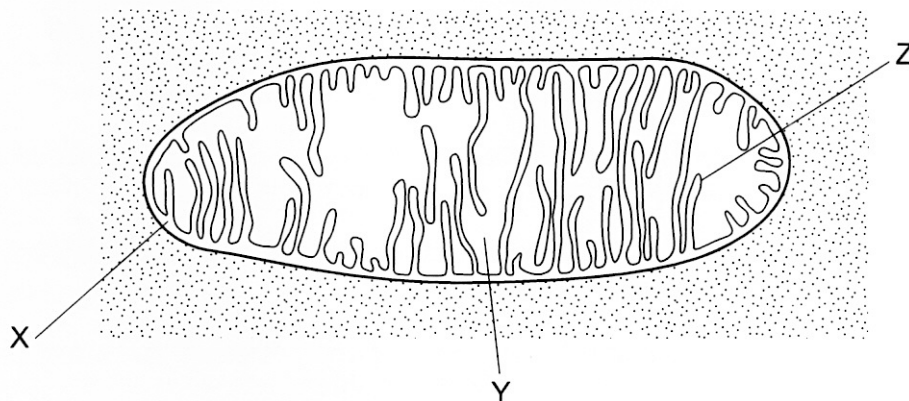
- 1 Wild type is dominant.
 - 2 The wild type parents are heterozygotes.
 - 3 Virescent is the recessive phenotype but the characteristic will only be expressed at lower temperature.
 - 4 The parents of cross 1 are only recessive at 20°C.
- A 2 only
 B 1 and 3 only
 C 1, 2 and 3 only
 D all of the above

- 22 The figure below shows 4 pedigrees. A shaded circle represents an affected female while a shaded square represents an affected male.



Which pedigree(s) show(s) a sex-linked dominant trait in humans?

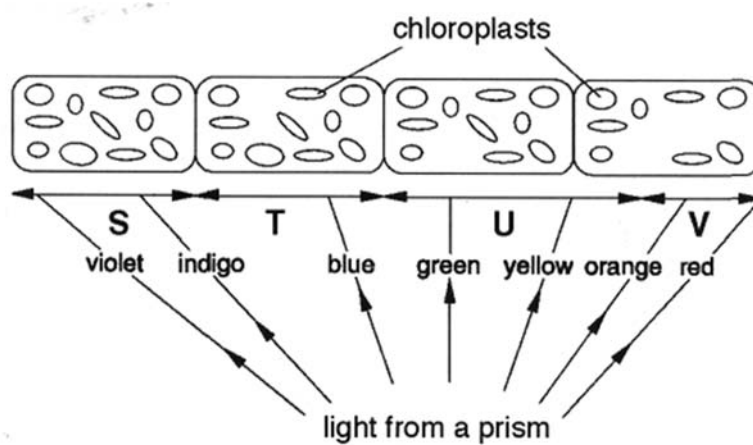
- A I only
 B I and III
 C II and III
 D III and IV
- 23 The electron micrograph shows an organelle found in a plant cell.



Which row shows the correct location of each named biological molecule?

	DNA	oxaloacetate	ATP
A	absent	Y	Z
B	Y	Z	Y
C	Y	Y	X
D	Y	Y	Y

- 24 A filament of the alga *Cladophora* was illuminated as shown in the diagram. Motile, aerobic bacteria were placed in the water with the alga and their position was determined by microscopic examination after 10 minutes.



In which two regions will the number of bacteria be highest?

- A S and T
 - B S and U
 - C T and V
 - D U and V
- 25 The copying of DNA is not always accurate, resulting in occasional, random changes to its base sequence. For example, scientists have calculated that one amino acid in vertebrate haemoglobin protein is changed every 10 million years.

Which statement about this process is correct?

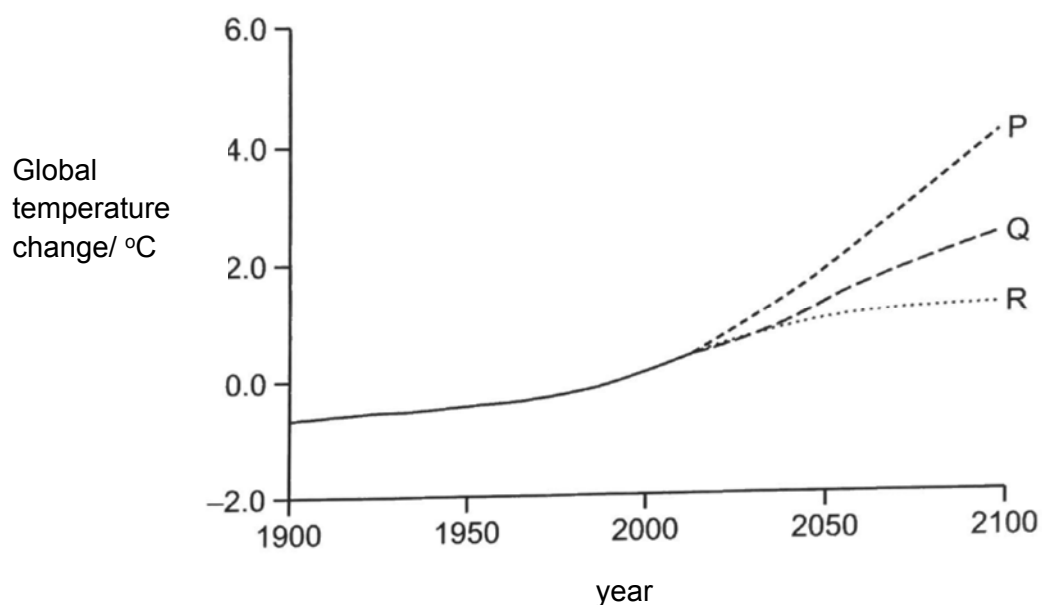
- A A single amino acid change in a protein may be affected by natural selection.
- B All changes in the base sequence of DNA are affected by natural selection.
- C Natural selection can cause some of the amino acid changes in proteins.
- D The variability of DNA in a population will be less than the variability of its proteins.

- 26** In 1954, copper waste in the Finniss River killed numerous fish. This caused various species in the area to die out. However, one species, the black-banded rainbow fish, increased in numbers. The black-banded rainbow fish have modified gills that enable the fish to filter and remove the copper before it enters their body.

With respect to the black-banded rainbow fish it is reasonable to conclude that

- A** a mutation occurred in their population in 1954.
 - B** the ability of their gills to remove copper already existed in 1954
 - C** the high levels of copper in the water changed the structure of their modified gills
 - D** their genomes are identical with those of the other species of fish that existed in 1954.
- 27** Climate change has led to increasing temperature. The environments where birds, fishes, and other marine species live have become warmer. What is not an effect of climate change on these organisms?
- A** Certain fish species migrate in response to seasonal temperature changes, moving northward or to deeper, cooler waters in the summer and migrating back during the winter.
 - B** The birds are moving north to feed and breed in the summer, then moving further south to spend the winter in warmer areas.
 - C** The birds follow a regular seasonal migration pattern during wintering periods.
 - D** As smaller marine prey species shift their habitats, larger predator species may follow them.
- 28** Other than melting land-based ice sheets, which of these factors has made the largest contribution to the rise in sea level over the past 100 years?
- A** Warming of ocean surface water
 - B** Melting sea ice
 - C** Increased river runoff
 - D** Increased reflection of infrared radiation

- 29 The graph shows the predicted changes in global temperatures using three different models, P, Q and R. Model Q assumes that no new factors act to influence the rate of climate change.



The predictions based on models P and R can be explained using some of the following statements.

- 1 An increased global temperature and reduced rainfall will lead to an increase in forest fires.
- 2 Permanently frozen soil and sediment in the Arctic will begin to thaw as global temperature increase.
- 3 Rising sea temperature will cause increase growth of photosynthetic algae.
- 4 Rising sea temperatures will reduce the solubility of greenhouse gases in the oceans.

Which of these statements support predictions P and R?

	statements that support prediction P	statements that support prediction R
A	1, 2 and 4	3
B	1 and 3	2 and 4
C	2	1, 3 and 4
D	3 and 4	1 and 2

30 Corals are among the first indicators of climate change. When ocean temperatures get too hot for too long, corals undergo a process called bleaching. Which statements are true about this process?

- 1 Increased levels of sediment in seawater, the zooxanthellae may lose substantial amounts of their photosynthetic pigmentation, which decreases rates of photosynthesis produces bleaching.
- 2 Bleaching occurs when abnormally high sea temperatures cause corals to expel the zooxanthellae living in them.
- 3 Zooxanthellae are able to use the oxygen and waste materials of the host, supplying carbon dioxide and food substances in return.
- 4 With high sea temperatures and decreasing planktons, corals use the zooxanthellae as their food source

- A** 1 and 2
- B** 1 and 4
- C** 2 and 4
- D** 2 and 3